

Draft 8 — Author Action Summary

The Face-Adjacency Graph of the Truncated Octahedron

Scientific recommendation: conditional minor revision. No confirmed fatal or major mathematical error was established in the examined submission. This is not a certification of the unexecuted tagged release.

1. **Complete the immutable computational deposit (required).** In Section 8, provide the Zenodo DOI or other immutable archive identifier, the exact commit behind `face-graph-note-v8`, pinned dependencies, the generated certificate and numeric/figure data. Test the documented one-command runner from a clean environment, including an intentional failure-path check. The present referee examination could not execute the complete tagged release; this is a documented limit rather than an allegation of program failure.
2. **Refine the prior-art positioning (minor).** Cite or distinguish MathWorld’s known graph/adjacency spectrum and Avishai–Luck’s spherical magnetic-charge spectral methods from the present graph-specific Laplacian and quarter-flux factorization. The search found no exact predecessor within its stated bounds; do not present that as a proof of novelty.
3. **Keep the mathematical hypotheses explicit (retain).** Proposition 6.1 is stated for a cellular embedding on S^2 . The sphere assumption is essential; a toroidal lattice permits distinct winding holonomies with identical face fluxes. The present graph’s 24 triangular boundaries span its 23-dimensional cycle space. No correction to the theorem was established.

Editorial reversal condition. A validated in-scope counterexample to the spherical gauge theorem, or a reproducible mismatch between the specified magnetic connection and the claimed quarter-flux factorization, would require major rereview and could reverse the favorable mathematical recommendation. An unremedied release-reproduction failure prevents final clearance. *The full referee report contains the mathematical reasoning, evidence bounds, individual metrics and references.*